
Unit – 1 : Session – 12 : SLO – 1 | System Programs vs System Calls — Diagram Keywords

I. Diagram Keywords (Layered View)

1. Top layer: USER PROGRAMS / APPLICATIONS
2. Next: SYSTEM PROGRAMS (utilities: shells, editors, compilers, loaders)
3. Next: SYSTEM CALL INTERFACE (API: open, read, write, fork, exec)
4. Core: KERNEL (process management, memory management, file system, device drivers)
5. Bottom: HARDWARE (CPU, memory, devices)

II. Distinction

1. SYSTEM CALLS: the programmatic interface between a process and the kernel (e.g. fork, read).
2. SYSTEM PROGRAMS (utilities): executable programs shipped with the OS that USE system calls to
3. provide user-facing functionality (e.g. ls, cp, gcc, top). They run in user mode, not kernel mode.